

Notes: Duflo and Pande (QJE, 2007)

Dams

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1 One-sentence summary

Duflo and Pande use river gradient as an instrument for dam construction in India and show that dams raise agricultural production and reduce rainfall vulnerability downstream, but impose large welfare costs on the districts where dams are built, increasing poverty and volatility.

2 Big picture

2.1 Research question

The paper asks whether large irrigation dams deliver the productivity and poverty-reduction benefits promised by governments and development agencies, and how those benefits and costs are distributed spatially between dam-site districts and downstream districts.

2.2 Why dams are a useful policy setting

Dams create a natural spatial asymmetry. Districts downstream of dams may benefit from regulated water flows and irrigation. Districts where dams are located bear displacement, submergence, construction disruption, salinity, waterlogging, and restrictions on upstream water use. This allows the authors to separate beneficiaries from likely losers.

2.3 Main empirical result

Dams increase agricultural production downstream and reduce the sensitivity of production to rainfall shocks. In contrast, the district where a dam is built experiences weak or insignificant production gains but higher volatility and higher rural poverty.

2.4 Distributional lesson

The benefits and costs of public infrastructure are not automatically compensated through markets or public institutions. Even in a case where losers are geographically identifiable, the losers appear not to be fully compensated.

2.5 Policy implication

Infrastructure evaluation should not stop at aggregate output. It should ask who benefits, who loses, and whether institutions compensate losers. The paper is therefore a development-economics study of both productivity and distribution.

3 Incremental contribution of the paper

3.1 Causal evidence on dams rather than descriptive debate

Before this paper, debates over large dams relied heavily on engineering claims, case studies, and advocacy evidence. The paper contributes systematic econometric evidence on the average productivity and poverty effects of large dams across Indian districts.

3.2 A spatial treatment framework

The paper distinguishes *own-district dams* from *upstream dams*. This distinction is central because the same infrastructure can benefit one place while harming another.

3.3 A geographic instrument for infrastructure placement

The paper uses river gradient as an instrument for dam placement. Steeper river gradients make a district more suitable for dam construction, but conditional on geography and fixed effects, should affect outcomes mainly through dams.

3.4 Linking production, risk, and welfare

The paper does not only estimate output effects. It connects dams to agricultural production, rainfall-risk exposure, irrigation, public goods, migration, poverty, the poverty gap, malarial incidence, and inequality.

3.5 Evidence on incomplete compensation

The paper shows that dam-site districts suffer welfare losses even though the beneficiaries and losers are spatially predictable. This contributes to the literature on the political economy of development projects and limits of state compensation.

4 Data and measurement

4.1 Geographic unit and sample

The main unit of observation is the Indian district. The paper combines district-year agricultural data, repeated household survey poverty data, and spatial data on rivers, elevation, and dams.

4.2 Dam measures

The dam data come from the ICOLD World Register of Dams. The two key measures are:

- **Number of dams in district:** dams physically located in the district.
- **Number of dams upstream to district:** dams located upstream that may regulate water flows to the district.

Interpretation: Own-district dams measure the likely cost side of dam construction. Upstream dams capture the main productivity and irrigation benefits.

4.3 River gradient as suitability

The instrument is based on the gradient of rivers within a district. The key categories are low gradient, medium gradient, and high gradient river districts. Dam construction is concentrated in moderate-gradient districts.

Interpretation: The instrument is based on engineering feasibility: river slope affects whether a location is suitable for dam construction. The exclusion restriction requires that, after controls, river gradient affects outcomes through dam construction rather than through other channels.

4.4 Agricultural outcomes

Agricultural outcomes include gross cultivated area, gross irrigated area, crop production, yield, and production volatility. The paper uses the monetized value of production for six major crops and measures output in logs where appropriate.

4.5 Welfare outcomes

Welfare outcomes include per-capita expenditure, rural headcount poverty, poverty gap, Gini coefficient, agricultural wage growth, and malaria incidence. These outcomes are observed from household surveys and public health data.

4.6 Rainfall shocks

Rainfall shocks measure deviations from normal rainfall. The authors use rainfall interactions to ask whether dams reduce vulnerability to bad rainfall shocks or amplify risk in the dam-site district.

5 Empirical specification and econometric design

5.1 Baseline district panel equation

The core estimating equation can be summarized as:

$$Y_{ist} = \alpha_i + \lambda_{st} + \beta_1 Dams_{ist} + \beta_2 UpstreamDams_{ist} + X'_{ist}\Gamma + \varepsilon_{ist},$$

where i indexes districts, s states, and t years. District fixed effects remove time-invariant district characteristics, while state-year fixed effects absorb shocks common to districts within the same state-year.

Interpretation: The comparison is within a state-year across districts that differ in exposure to own and upstream dam construction.

5.2 Instrumental variables

The endogenous dam variables are instrumented using interactions of river gradient categories with the number of dams built in the state-year. The first stage exploits the idea that when a state builds more dams, they are more likely to be built in districts with suitable river gradients.

$$Dams_{ist} = \pi_1(StateDams_{st} \times RiverGradient_i) + controls + u_{ist}.$$

Interpretation: The IV shifts dam construction toward geographically suitable districts, separating the effect of dams from the possibility that governments build dams in districts already trending differently.

5.3 Downstream exposure

For each district, the authors compute the number of dams upstream using river-flow information. This separates the effects of being the construction site from receiving downstream water regulation benefits.

5.4 Rainfall-risk specification

To test risk mediation, the paper interacts dams with rainfall shocks:

$$Y_{ist} = \dots + \theta_1 RainShock_{it} + \theta_2(Dams_{ist} \times RainShock_{it}) + \theta_3(UpstreamDams_{ist} \times RainShock_{it}) + \varepsilon_{ist}.$$

Interpretation: A negative upstream interaction for poverty or a positive upstream interaction for production would indicate that upstream dams buffer districts against rainfall risk. A harmful own-district interaction indicates increased vulnerability near the dam.

5.5 Institutional heterogeneity

The paper tests whether welfare costs are lower in districts with stronger local institutional capacity, using historical land revenue institutions and tribal population shares.

Interpretation: This asks whether institutions can mediate compensation. If redistribution works better in some places, dam-site poverty costs should be smaller there.

6 Identification and interpretation

6.1 What identifies the effect of dams

The paper identifies dam effects from variation in dam construction induced by river-gradient suitability within state-years. Conditional on district fixed effects, state-year fixed effects, and geography controls, river-gradient interactions provide plausibly exogenous variation in dam exposure.

6.2 Key exclusion restriction

The instrument must affect outcomes through dam construction, not through independent effects of river gradient. The paper addresses this by controlling flexibly for geography, including elevation, river length, district area, and gradient-year interactions.

6.3 Why own-district and upstream effects differ

The technology of dams produces spatially opposite effects. A district where a dam is built may lose land and face displacement. Downstream districts receive regulated water and irrigation benefits. Estimating both effects is therefore essential.

6.4 Interpretation of poverty results

The poverty increase in dam-site districts is not a mere output result. It indicates that construction costs, displacement, land loss, disease exposure, or weak compensation dominate local gains. The downstream poverty reductions show that benefits exist, but not for the same people.

6.5 Main limitations

The paper cannot observe all displaced households or perfectly measure compensation. It also cannot fully distinguish displacement, salinity, disease, and water-use restrictions as separate channels. The estimates are district-level average effects, not household-level treatment effects.

7 Tables and figures

7.1 Table I: Descriptive Statistics

Read: Table I summarizes geography, dam exposure, welfare, agriculture, demographics, and public goods at the beginning and end of the sample. It shows that the average number of dams in a district rises from 2.39 to 8.66, while upstream dams rise from 3.63 to 13.85.

Economic message: The table establishes both the scale of dam construction and the large changes in poverty, irrigation, production, and public goods over the sample.

Interpretation: The descriptive statistics do not identify effects, but they show why dam construction is quantitatively important and why controls are necessary.

	Beginning period	End period	Source
<i>A. Geography</i>			
Fraction district with river gradient 0-1.5%	0.75 (0.30)		GIS
Fraction district with river gradient 1.5-3%	0.08 (0.07)		GIS
Fraction district with river gradient 3-6%	0.06 (0.08)		GIS
Fraction district with river gradient above 6%	0.11 (0.22)		GIS
<i>B. Dams</i>			
Number of dams in district	2.39 (4.53)	8.66 (16.98)	ICOLD Dam Register
Number of dams upstream to district	3.63 (7.77)	13.85 (30.57)	ICOLD Dam Register
<i>C. Welfare</i>			
Per capita expenditure (log Rupees)	3.80 (0.20)	5.79 (0.27)	National Sample Survey
Headcount ratio	0.46 (0.16)	0.22 (0.14)	National Sample Survey
Poverty gap	0.27 (0.05)	0.04 (0.04)	National Sample Survey
Malaria incidence	1.31 (2.10)	0.47 (1.85)	National Malaria Eradication program
Agricultural wage (log Rupees)	1.44 (0.35)	1.71 (0.31)	Evenson-McKinsey
<i>D. Agriculture</i>			
Gross cultivated area (in '000 hectares)	570 (277)	642 (324)	Evenson-McKinsey
Gross irrigated area (in '000 hectares)	138 (151)	169 (176)	Evenson-McKinsey
Total production (Rupees per '000 tons)	15,931 (12,749)	81,336 (110,130)	Evenson-McKinsey and own update
Yield (Rupees per hectare)	52.26 (30)	115.27 (92)	
Rainfall shock	0.079 (0.25)	-0.231 (0.23)	Delaware rainfall data

	Beginning period	End period	Source
<i>Geography and Public Goods</i>			
Population (in '000 persons)	1,295 (867)	1,887 (1,254)	Census of India
In-migrants (in '000 persons)	396 (239)	486 (309)	Census of India
Population villages in district with any water facility	0.982 (0.03)	0.997 (0.01)	Census of India
Population villages in district with power	0.24 (0.26)	0.81 (0.22)	Census of India
Population villages in district with tarmac road	0.31 (0.19)	0.47 (0.26)	Census of India
Landlord districts (proportion)		0.61 (0.48)	Banerjee and Iyer (2005)
Population share in district in 1971		0.10 (0.20)	Census of India [1971]

Beginning and end periods are (i) dams: 1971 and 1999; (ii) welfare measures: 1973 and 1999 (for malaria 1971 and 1999); (iii) production and rainfall: 1971 and 1999; (iv) area and 71 and 1987; (v) demographic and public goods: 1971 and 1991. All monetary values are in 1973 Rupees (at current exchange rate 44 Rupees = 1 U.S. dollar). Standard deviations below. Total production includes production of six major crops: wheat, rice, sorghum (jowar), sugarcane, pearl millet (bajra), maize. Headcount ratio is the proportion population above the poverty line. Poverty gap is the average distance below the line expressed as a proportion of the poverty line (the average is over the whole population with the non-poor distance below the line). Malaria incidence is the log of average parasite incidence in the district. Rainfall shock is the fractional deviation of districts rainfall from the district or 1971-1999. Non-landlord district is a district-level dummy which equals one if majority area in the district was under non-landlord arrangement. Further details on the data and variable construction are in the Appendix.

(a) Table I, part 1: geography, dams, welfare, and agricultural variables.

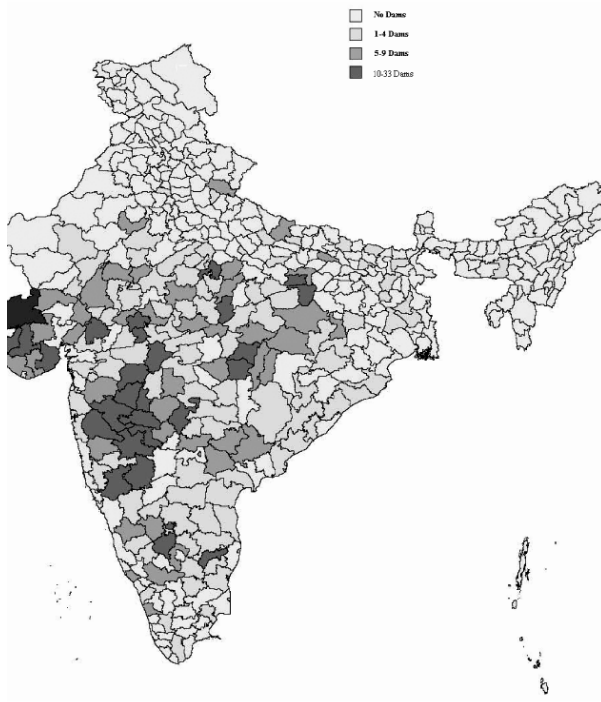
(b) Table I, part 2: demographics and public goods.

7.2 Figure I: Distribution of Dams across Indian Districts, 1970; Figure II: Distribution of Dams across Indian Districts, 1999

Read: Figure I maps dams in 1970; Figure II maps dams in 1999. Dam construction expands dramatically and remains geographically concentrated, especially in western and peninsular India.

Economic message: The figures motivate the need for a geography-based identification strategy. Dams are not randomly scattered across India; they are tied to terrain and river networks.

Interpretation: The maps show why naive comparisons could be biased: dam-rich districts differ geographically from dam-poor districts. The IV strategy turns that geographic pattern into a source of exogenous variation.



(a) Figure I: distribution of dams in 1970.

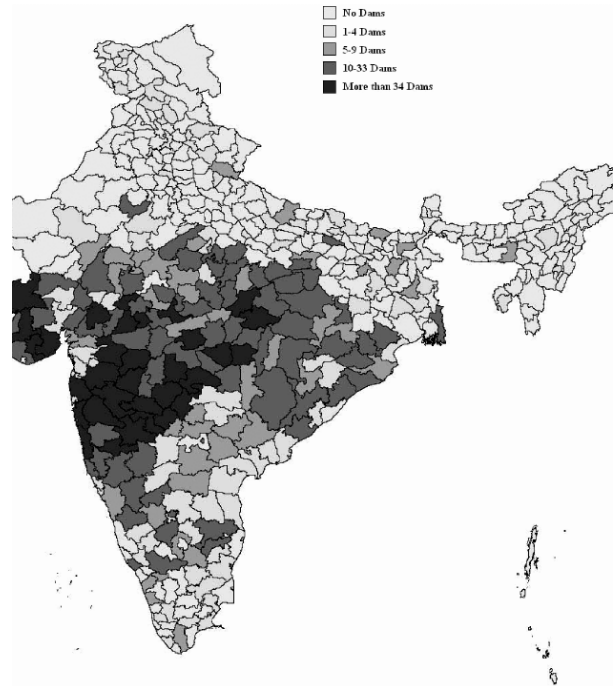


FIGURE II
The Distribution of Dams across Indian Districts, 1999

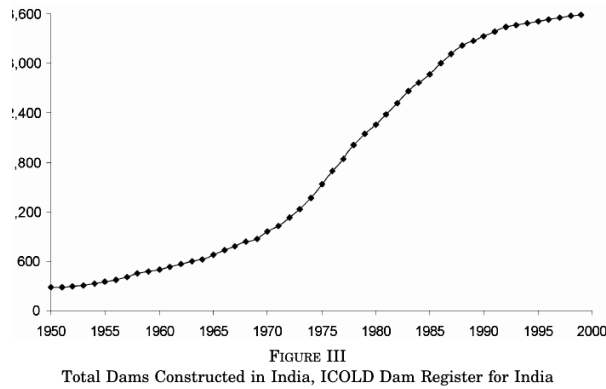
(b) Figure II: distribution of dams in 1999.

7.3 Figure III: Total Dams Constructed in India, ICOLD Dam Register for India; Figure IV: Average River Gradient by District

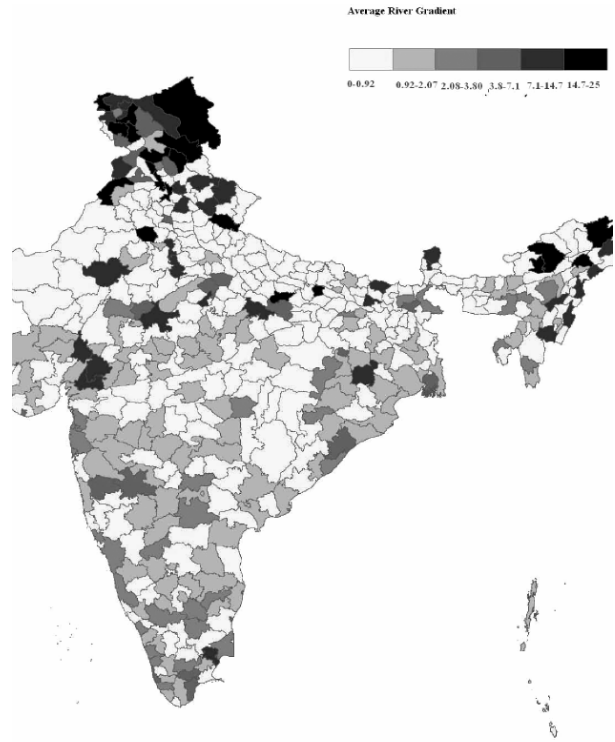
Read: Figure III shows the national time series of completed dams, with a rapid increase after independence. Figure IV maps river gradients by district, the geographic basis of the instrument.

Economic message: Together, these figures show the two components of the first stage: time variation in dam construction and cross-sectional variation in suitability.

Interpretation: Figure III provides the timing of treatment expansion, while Figure IV provides the cross-sectional assignment mechanism.



(a) Figure III: total dams constructed over time.



(b) Figure IV: average river gradient by district.

7.4 Table II: Geography and Dam Construction; Table III: Dams and Agriculture

Read: Table II shows that river gradient predicts dam construction. Table III shows that upstream dams increase agricultural production, irrigated area, and production stability, while own-district dams have weaker or adverse effects.

Economic message: The first table is the first stage; the second table is the main production result. Downstream districts benefit from water control, while dam-site districts do not receive the same production gains.

Interpretation: The positive upstream effect is the central evidence that dams can raise agricultural productivity. The weak own-district effect suggests that construction costs and local disruption offset benefits near the dam.

TABLE II
GEOGRAPHY AND DAM CONSTRUCTION

	Number of dams		
	Cross-section (1999)	Poverty sample	Production sample
	Not interacted	Interacted with predicted number of dams in the state	
	(1)	(2)	(3)
Fraction river gradient	0.278	0.153	0.176
1.5–3%	(0.122)	(0.040)	(0.094)
Fraction river gradient	−0.210	−0.191	−0.219
3–6%	(0.127)	(0.065)	(0.128)
Fraction river gradient	0.014	0.075	0.097
above 6%	(0.033)	(0.031)	(0.043)
F-test for river gradient	1.764	6.372	7.68
[p-value]	[0.15]	[0.000]	[0.053]
Geography controls	Yes	Yes	Yes
State*year and river gradient*year interactions	No	Yes	Yes
Fixed effects	State	District	District
N	374	1855	7743

Regressions include district fixed effects, state*year interactions, interaction of the number of predicted dams in the state with district gradient, kilometers of river, district area, and district elevation and river gradient*year interactions (see notes to Table II for a full description of geography variables). They also include interaction of the number of predicted dams in the state with (average) gradient, kilometers of river, area and elevation in upstream districts, river gradient in upstream districts*year interaction, and an indicator for whether the district has any upstream districts. Regression coefficients are multiplied by 100. Standard errors, clustered by district, are reported in parentheses. Production and yield variables are in kg/ha. We use the annualized value of production for an average crop (rice) in Table I. Yield is defined as crop production per unit of land (ha, per hectare). Nonwater-intensive crops are sorghum (lower), pearl millet (upper), and maize, and water-intensive crops are wheat, rice, and sugarcane. The sample includes annual data for 271 districts in 13 states (defined by 1982 census boundaries). Area and fertilizer data cover 1971–1987 and production and yield data 1971–1989. Deviations in sample size are due to missing data. The last row provides the F-statistic from the regression of the number of dams in the district on the predicted number of dams in the own district.

(a) Table II: geography and dam construction.

	TABLE III DAMS AND AGRICULTURE								
	Area				Inputs	Agricultural production			
	Gross irrigated area		Gross cultivated area		Fertilizer use	Production	Yield	Production	
	Level	Log	Level	Log		All crops		Water-intensive crops	Non-water-intensive crops
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Part A. FGLS</i>									
Dams									
Own district	14.528	0.131	114.493	0.094	0.231	0.184	0.152	0.063	0.640
	(13.300)	(0.156)	(47.838)	(0.059)	(0.342)	(0.234)	(0.196)	(0.234)	(0.585)
Upstream	17.830	0.298	77.641	0.028	0.256	0.227	0.227	0.269	0.801
	(12.639)	(0.162)	(48.233)	(0.054)	(0.339)	(0.155)	(0.141)	(0.243)	(0.307)
<i>Part B. Feasible Optimal IV</i>									
Dams									
Own district	232.092	0.728	325.358	0.875	0.563	0.085	−0.033	0.366	−0.105
	(235.847)	(1.002)	(263.509)	(0.500)	(1.244)	(0.099)	(0.451)	(0.782)	(1.349)
Upstream	49.754	0.325	58.602	0.096	0.169	0.341	0.193	0.470	0.181
	(22.339)	(0.154)	(35.674)	(0.062)	(0.175)	(0.118)	(0.097)	(0.154)	(0.307)
N	4,638	4,638	4,622	4,622	4,621	7,078	7,077	7,143	6,766
First stage	8.48	8.48	8.51	8.51	8.5	9.22	9.22	9.03	9.14
F-statistic (own district)									

Regressions include district fixed effects, state*year interactions, interaction of the number of predicted dams in the state with district gradient, kilometers of river, district area, and district elevation and river gradient*year interactions (see notes to Table II for a full description of geography variables). They also include interaction of the number of predicted dams in the state with (average) gradient, kilometers of river, area and elevation in upstream districts, river gradient in upstream districts*year interaction, and an indicator for whether the district has any upstream districts. Regression coefficients are multiplied by 100. Standard errors, clustered by district, are reported in parentheses. Production and yield variables are in kg/ha. We use the annualized value of production for an average crop (rice) in Table I. Yield is defined as crop production per unit of land (ha, per hectare). Nonwater-intensive crops are sorghum (lower), pearl millet (upper), and maize, and water-intensive crops are wheat, rice, and sugarcane. The sample includes annual data for 271 districts in 13 states (defined by 1982 census boundaries). Area and fertilizer data cover 1971–1987 and production and yield data 1971–1989. Deviations in sample size are due to missing data. The last row provides the F-statistic from the regression of the number of dams in the district on the predicted number of dams in the own district.

(b) Table III: dams and agriculture.

7.5 Table IV: The Reach of Dams: 2SLS Estimates; Table V: The Impact of Dams: Robustness Check

Read: Table IV examines how the effects vary with alternative upstream/downstream definitions. Table V shows that the production and poverty results are robust to additional controls and alternative samples.

Economic message: These tables check that the main conclusions are not driven by one arbitrary definition of a dam’s reach or by omitted controls.

Interpretation: The central pattern survives: downstream exposure improves production and welfare, while own-district exposure raises poverty.

TABLE IV
THE REACH OF DAMS: 2SLS ESTIMATES

	Agricultural production	Headcount ratio	Poverty gap
	(1)	(2)	(3)
<i>Part A. Neighboring districts</i>			
Dams			
Own district	0.345	0.594	0.223
	(1.148)	(0.306)	(0.101)
Upstream	0.665	−0.170	−0.045
	(0.220)	(0.078)	(0.023)
Downstream	−0.107	0.073	0.022
	(0.260)	(0.060)	(0.021)
Neighboring but not upstream/downstream	−0.175	0.013	−0.008
	(0.203)	(0.072)	(0.026)
N	7,078	1,799	1,799
<i>Part B. Time Effects</i>			
Dams			
Own district	0.109	0.888	0.417
	(1.266)	(0.444)	(0.194)
Own district, completed in next 5 years	0.891	2.004	0.963
	(3.376)	(1.213)	(0.433)
Upstream	0.187	−0.156	−0.049
	(0.323)	(0.154)	(0.058)
Upstream, completed in next 5 years	−1.127	0.061	−0.025
	(0.683)	(0.372)	(0.137)
N	4,992	1,443	1,443

Regressions include district fixed effects, state*year interactions, interaction of predicted dams in the state with district gradient, kilometers of river, district area, and district elevation and river gradient*year interactions (see Table II for a description of the geography variables). They also include river gradient in upstream districts*year interactions and interaction of predicted dams in the state with (average) gradient, kilometers of river, area and elevation in upstream districts, and an indicator for whether the district has any

(a) Table IV: reach of dams, 2SLS estimates.

	TABLE V THE IMPACT OF DAMS: ROBUSTNESS CHECK							
	Agricultural production				Headcount ratio			
	Additional control				Additional control			
	Initial poverty	Tribal population	Linear trend	Dams per 100 sq. km	Initial poverty	Tribal population	Linear trend	Dams per 100 sq. km
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Part A. OLS / FGLS</i>								
Dams								
Own district	0.168	0.207	0.144	0.168	0.279	0.266	0.250	0.310
	(0.310)	(0.299)	(0.308)	(0.310)	(0.074)	(0.084)	(0.086)	(0.100)
Upstream	0.498	0.488	0.470	0.498	−0.077	−0.076	−0.084	−0.086
	(0.137)	(0.140)	(0.134)	(0.137)	(0.030)	(0.041)	(0.039)	(0.033)
<i>Part B. 2SLS / Feasible Optimal IV</i>								
Dams								
Own district	0.082	−0.162	0.219	0.082	0.748	0.782	0.256	0.867
	(0.769)	(0.676)	(1.729)	(0.769)	(0.308)	(0.296)	(0.765)	(0.427)
Upstream	0.379	0.269	0.363	0.379	−0.168	−0.143	−0.185	−0.150
	(0.115)	(0.115)	(0.163)	(0.115)	(0.068)	(0.069)	(0.076)	(0.071)
N	7,078	7,078	7,078	7,078	1,889	1,799	1,799	1,799

All regressions include district fixed effects, state*year interactions, interaction of the number of predicted dams in the state with district gradient, kilometers of river, district area, and district elevation and river gradient*year interactions (see notes to Table II for a full description of geography variables). They also include interaction of the number of predicted dams in the state with (average) gradient, kilometers of river, area and elevation in upstream districts, river gradient in upstream districts*year interaction, and an indicator for whether the district has any upstream districts. The additional controls are in columns (1) and (6) the separate interactions of the number of predicted dams in the state with the 1971 tribal population share in own district and in upstream districts, and (3) in columns (2) and (5) the separate interactions of the number of predicted dams in the state with the 1971 tribal population share in own district and in upstream districts, and (4) in columns (3) and (6) 1975 state dam share*linear time trend interacted with river gradient in own district and upstream districts. Regression coefficients in columns (1)–(3) and (5)–(7) are multiplied by 100. Coefficients in columns (4) and (8) are per 100 square kilometers. Standard errors are in parentheses. Columns (1)–(3) report OLS; Part A and Feasible Optimal IV. Part B estimates with standard errors clustered by district. Columns (4)–(8) report OLS/Part A and 2SLS/Part B) estimates with standard errors clustered by NSS region*year.

(b) Table V: robustness checks.

7.6 Table VI: Dams and Rainfall Shocks; Table VII: Dams and Public Good Provision and Demographics

Read: Table VI shows how dams change the effect of rainfall shocks on production and poverty. Table VII asks whether dams change water facilities, power, roads, population, and migration.

Economic message: Upstream dams reduce vulnerability to rainfall shocks, while own-district dams can amplify adverse shocks. Table VII checks whether public goods or migration explain the welfare patterns.

Interpretation: The rainfall table provides a mechanism: dams are valuable partly because they smooth water supply downstream. The public goods table suggests limited evidence that infrastructure or migration fully offsets local costs.

TABLE VI
DAMS AND RAINFALL SHOCKS

	Agricultural production		Headcount ratio		Poverty gap	
	(1)	(2)	(3)	(4)	(5)	(6)
Rainshock	0.065 (0.030)	0.008 (0.044)	-0.041 (0.019)	-0.036 (0.026)	-0.012 (0.007)	-0.0001 (0.010)
Dams	-0.011 (1.227)	0.109 (1.228)	0.765 (0.324)	0.713 (0.320)	0.294 (0.111)	0.255 (0.108)
Dams*rainshock		0.898 (0.364)		-0.243 (0.191)		-0.203 (0.086)
Upstream dams	0.722 (0.197)	0.734 (0.195)	-0.149 (0.067)	-0.130 (0.068)	-0.038 (0.019)	-0.030 (0.019)
Upstream dams*rainshock		-0.134 (0.092)		0.109 (0.059)		0.034 (0.022)
N	7,078	7,078	1,799	1,799	1,799	1,799

All columns report 2SLS regressions. Regressions include district fixed effects, state*year interactions, interaction of the number of predicted dams in the state with district gradient, kilometers of river, district area, and district elevation and river gradient*year interactions (see notes to Table II for a full description of geography variables). They also include interaction of the number of predicted dams in the state with (average) gradient, kilometers of river, area and elevation in upstream districts, river gradient in upstream districts*year interaction, and an indicator for whether the district has any upstream districts. All coefficients are multiplied by 100. Standard errors reported in parentheses. These are clustered by district in columns (1)-(2), and by 250 regions*year in columns (3)-(6). Rainshock is the fractional annual deviation of district rainfall from its historic mean (defined over 1971-1999). Agricultural production is in logs. We have annual agricultural data for 271 districts and the years 1971-1999, and poverty data for 374 districts and 1971, 1981, 1985, 1991, and 1999. Missing district*year observations account for actual sample size.

TABLE VII
DAMS AND PUBLIC GOOD PROVISION AND DEMOGRAPHICS: 2SLS ESTIMATES

	Water facility	Power	Tarmac road	Population	In-migrants
	(1)	(2)	(3)	(4)	(5)
<i>Dams</i>					
Own districts	0.090 (0.058)	0.034 (0.347)	-0.070 (0.240)	0.082 (0.574)	-0.293 (0.797)
Upstream	-0.014 (0.011)	0.157 (0.093)	0.059 (0.065)	0.057 (0.059)	0.160 (0.104)
N	847	848	848	947	947

Regressions include district fixed effects, state*year interactions, river gradient*year interactions, and interaction of number dams in the state with (i) district gradient variables, (ii) kilometers of river and district area, and (iii) elevation variables and river gradient*year interactions (see Table II for a full description of the geography variables). They also include interaction of the number of dams in the state with (average) gradient, kilometers of river, area and elevation in upstream districts, river gradient in upstream districts*year interactions, and an indicator for whether the district has any upstream districts. Regression coefficients are multiplied by 100, and robust standard errors are in parentheses. The dependent variables in columns (1)-(3) are the fraction of villages in the district with the specified public good facility. Water facility excludes dams. The dependent variables in columns (4)-(5) are in logs and refer to the rural population. The unit of observation is the Indian district, and we have data for 1971, 1981 and 1991. Columns (1)-(3) regressions use data for 302 districts, as defined by 1971 census, and columns (4)-(5) regressions data for 339 districts as defined by the 1961 census. Differences in sample size are accounted for by missing district*year observations.

(a) Table VI: dams and rainfall shocks.

(b) Table VII: public goods and demographics.

7.7 Table VIII: Dams and Rural Welfare; Table IX: Institutions and Dams: 2SLS Estimates

Read: Table VIII reports the main rural welfare results. Own-district dams increase poverty, while upstream dams tend to reduce poverty. Table IX asks whether institutional variables mediate the effects.

Economic message: The welfare effects reveal the distributional tradeoff: aggregate productivity gains downstream coexist with welfare losses in dam-site districts.

Interpretation: Table VIII is the main distributional result. Table IX explores whether institutions reduce the burden, but the evidence does not overturn the basic conclusion that compensation is incomplete.

	Per-capita expenditure	Headcount ratio			Poverty gap	Gini coefficient	Agricultural wage growth	Malaria incidence
		Original	Assume poor in-migrants	Assume rich in-migrants				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Part A. OLS/FGLS</i>								
Dams								
Own district	-0.289 (0.115)	0.273 (0.084)	0.407 (0.083)	0.174 (0.081)	0.081 (0.030)	0.014 (0.027)	0.018 (0.043)	0.60 (0.89)
Upstream	0.083 (0.057)	-0.083 (0.038)	-0.079 (0.038)	-0.082 (0.038)	-0.027 (0.013)	0.007 (0.013)	0.057 (0.020)	0.20 (0.65)
<i>Part B. 2SLS/Flexible Optimal IV</i>								
Dams								
Own district	-0.457 (0.467)	0.772 (0.324)	0.879 (0.314)	0.651 (0.315)	0.297 (0.112)	0.104 (0.139)	-0.045 (0.242)	-0.40 (2.02)
Upstream	0.142 (0.084)	-0.154 (0.068)	-0.148 (0.066)	-0.150 (0.066)	-0.039 (0.020)	0.000 (0.020)	0.009 (0.031)	0.07 (0.52)
N	1,799	1,799	1,799	1,799	1,799	1,799	4,272	7,001
First stage F-statistic (own district)	7.71	7.71	7.71	7.71	7.71	7.71	8.71	10.90

Regressions include district fixed effects, state*year interactions, river gradient*year interactions, and interaction of number dams in the state with (i) district gradient variables, (ii) kilometers of river and district area, and (iii) elevation variables (see Table II for a full description of the geography variables). They also include interaction of the number of dams in the state with (average) gradient, kilometers of river, area and elevation in upstream districts, river gradient in upstream districts*year interactions, and an indicator for whether the district has any upstream districts. Regression coefficients are multiplied by 100. Column (1) OLS regression report (see Part A and IV Part B) estimates. Standard errors, clustered by 1,973 NSS region*year, are in parentheses. The unit of analysis is the Indian district as defined by 1981 census, we have data for 274 districts and were 1971, 1981, 1991, and 1999. Column (7) regression report over 1971-1997 for 274 districts (as defined by 1981 census). Column (8) regression report over 1971-1997 for 415 districts (defined by 1981 census). In both columns we report FGLS (Part A) and Flexible Optimal IV (Part B) estimates with standard errors clustered by district. Per-capita expenditure is in logarithms. Agricultural wage growth is defined as $\log(\text{wage}_{it}) - \log(\text{wage}_{it-1})$. Column (2) uses head-count ratio figures as computed from NSS data. Column (3) and (4) adjust this using the Table V in-migration estimates. Column (5) assumes in-migrants are poor, and column (6) in-migrants are rich. Malaria incidence is measured as log annual Parasite Incidence (API) = $\log(\text{No. of blood smears found positive for malarial/total population under surveillance})$.

TABLE IX
INSTITUTIONS AND DAMS: 2SLS ESTIMATES

	Agricultural production	Headcount ratio	Poverty gap
	(1)	(2)	(3)
Dams	-0.439 (2.129)	1.072 (0.539)	0.332 (0.178)
Dams*nonlandlord dummy	-0.125 (1.067)	-0.639 (0.309)	-0.193 (0.102)
Dams*tribal population share	1.354 (2.637)	0.711 (0.790)	0.087 (0.276)
Upstream dams	1.015 (0.708)	-0.393 (0.293)	-0.131 (0.102)
Upstream dams*Nonlandlord dummy	0.160 (0.705)	0.196 (0.264)	0.061 (0.085)
Upstream dams*tribal population share	-0.609 (0.701)	-0.057 (0.191)	-0.015 (0.060)
N	4,090	914	914

The regressions include district fixed effects, state*year interactions, river gradient*year interactions, and interaction of (i) district gradient variables, (ii) kilometers of river and district area, and (iii) elevation variables (see Table II for a full description of the geography variables) with number of dams in the state (we use the predicted number in column (1) and actual number in columns (2) and (3)). They also include interaction of the number of dams in the state with (average) gradient, kilometers of river, area and elevation in upstream districts, an indicator for whether the district has any upstream districts and river gradient in upstream districts*year interactions. In column (1) we use predicted number of dams in a state. Standard errors in column (1) are clustered by district and in columns (2) and (3) by NSS region*year. The sample is restricted to the 151 districts under British direct rule. The nonlandlord dummy is a district-level dummy which equals one if the majority of land in the district was under nonlandlord arrangements for land taxation purposes (from Banerjee and Iyer (2005)). The tribal population share is the fraction of district population that belongs to a tribal group, as measured in 1971. All coefficients are multiplied by 100.

(a) Table VIII: dams and rural welfare.

(b) Table IX: institutions and dams.

8 Short summary

- Dams increase agricultural production and reduce rainfall vulnerability downstream.
- Dam-site districts experience weak production gains and higher rural poverty.
- River gradient provides an instrument for dam construction because dam suitability depends on geography.
- The same project can generate gains in one location and losses in another.
- Institutions and markets do not appear to fully compensate dam-site losers.
- The paper is a major contribution to empirical development economics because it evaluates both productivity and distributional consequences of infrastructure.

9 Conclusion

9.1 Core conclusion

Large dams in India deliver agricultural gains, especially downstream, but they impose substantial welfare costs on the districts where they are built. The distributional costs are not fully offset by compensation, migration, public goods, or local institutions.

9.2 Economic intuition

The technology of dams separates costs from benefits. Reservoir districts lose land, face displacement, and may experience greater agricultural volatility. Downstream districts receive irrigation and water regulation. This spatial mismatch creates a distributional problem that standard cost-benefit analysis can hide.

9.3 Incremental contribution revisited

The paper contributes a credible empirical design for evaluating large infrastructure projects and shows that productivity gains and poverty effects can have opposite signs in different locations. It brings together

GIS geography, dam registries, agricultural data, rainfall data, and household poverty surveys in a unified evaluation.

9.4 Policy implication

Infrastructure projects should be evaluated with explicit attention to losers, not only aggregate output. If compensation fails in a setting where losers are geographically concentrated, compensation may be even harder in more diffuse policy settings.

9.5 Limitations

The paper cannot fully observe all displacement, compensation, or household-level adjustment. It also cannot separately quantify all mechanisms, such as land loss, waterlogging, malaria, or restrictions on upstream water use.

9.6 Takeaway

The main lesson is that development projects can simultaneously increase production and increase poverty for directly affected communities. Public infrastructure policy must therefore be judged by both efficiency and distribution.